

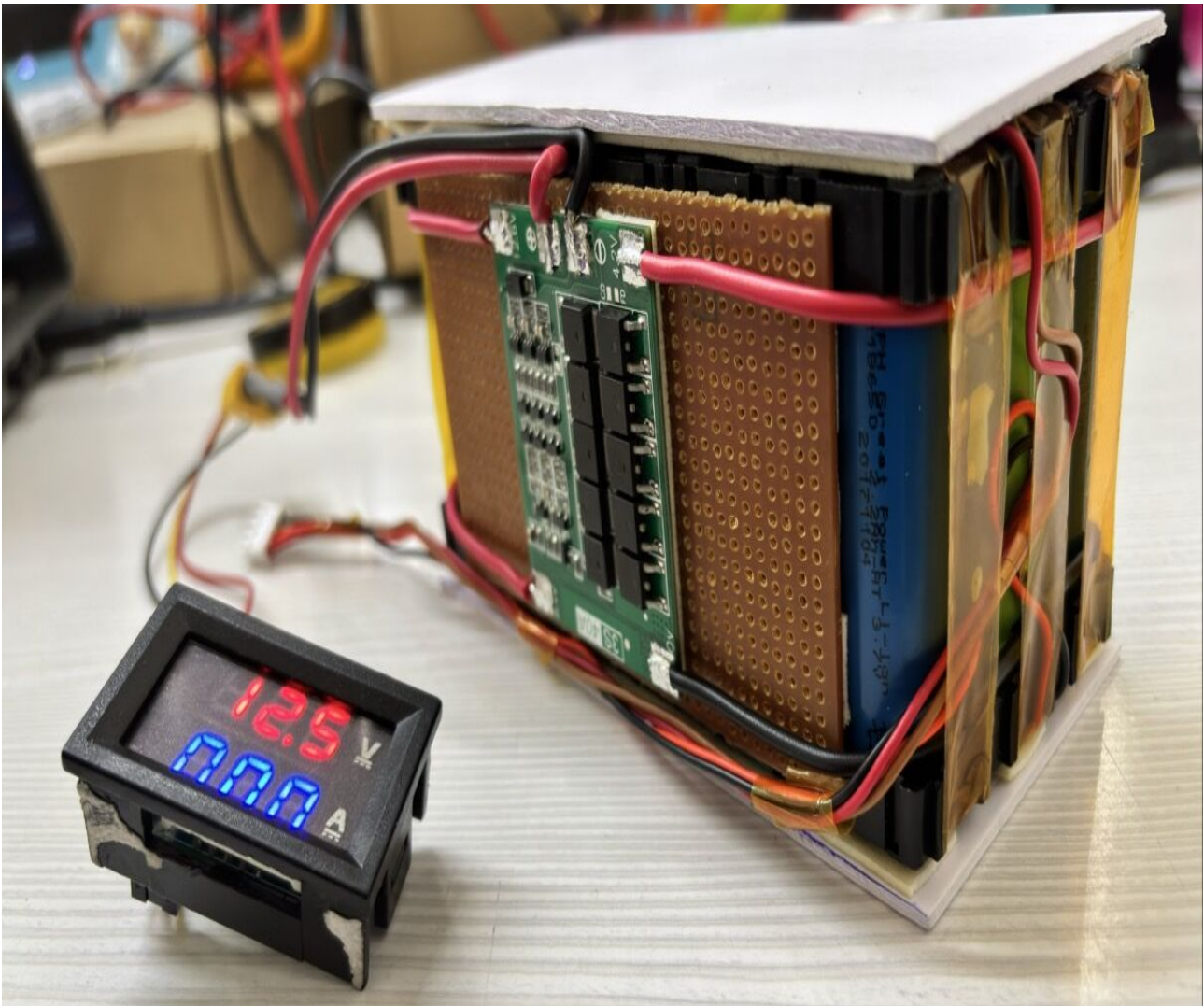
Test & Validation Report

3S6P Lithium-Ion Battery Pack — 12 Ah / 133.2 Wh

Project: 3S6P Li-Ion Battery Pack Design

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This report consolidates the physical hardware observations and the calculated electrical performance of the completed 3S6P battery-pack design. All numerical performance values below are either directly observed from the provided hardware photographs or calculated from the defined 20 m Ω pack model.



Completed physical battery pack with 3S BMS, balance wiring, XT60 output and voltage/current meter.

1. Test Objectives

- Verify the 3S6P electrical configuration and rated pack parameters.
- Verify the physical integration of the 3S BMS, balance connector and main output connection.
- Confirm the observed no-load pack voltage and current indication.
- Evaluate voltage drop, resistive power loss and efficiency using the defined 20 m Ω electrical model.

- Compare the calculated 0–30 A performance with the design targets.

2. Battery Under Test

Parameter	Validated / Calculated Result
Cell type	18650 Li-ion, 3.7 V nominal, 2.0 Ah
Configuration	3S6P
Total cells	18
Nominal pack voltage	11.1 V
Maximum pack voltage	12.6 V
Pack capacity	12 Ah
Nominal energy	133.2 Wh
Pack resistance model	20 mΩ
BMS	3S 40 A protection and balancing BMS
Maximum analyzed load	30 A

3. Hardware Verification

Verification item	Result	Validation basis
3S6P cell arrangement	18 cells / 3 series groups × 6 parallel	Design configuration
BMS integration	3S 40 A BMS installed	Physical inspection
Balance connector	4-pin balance connection present	Physical inspection
Main power connector	XT60 connector installed	Physical inspection
Panel meter	Voltage/current display installed	Physical inspection
Mechanical enclosure	Cell pack mechanically enclosed and insulated	Physical inspection
No-load voltage	≈12.5 V	Displayed hardware reading
No-load current	0.00 A	Displayed hardware reading

Observed hardware result: The photographed pack displays approximately 12.5 V and 0.00 A. A 12.5 V reading is consistent with a 3S lithium-ion pack near the upper part of its state-of-charge range and below the 12.6 V maximum-charge reference.

4. Electrical Performance Validation

The electrical performance is calculated using the project's defined pack resistance of 20 mΩ. The calculations use the following equations:

Voltage drop: $V_{\text{drop}} = I \times R$
Resistive loss: $P_{\text{loss}} = I^2 \times R$
Loaded voltage: $V_{\text{load}} = 11.1 - V_{\text{drop}}$
Load power: $P_{\text{load}} = V_{\text{load}} \times I$
Efficiency: $\eta = P_{\text{load}} / (11.1 \times I) \times 100$

Current	Voltage Drop	Loaded Voltage	Load Power	I ² R Loss	Efficiency
0 A	0.000 V	11.100 V	0.00 W	0.00 W	100.0%*
1.5 A	0.030 V	11.070 V	16.61 W	0.04 W	99.7%
3.0 A	0.060 V	11.040 V	33.12 W	0.18 W	99.5%
4.5 A	0.090 V	11.010 V	49.55 W	0.41 W	99.2%
6.0 A	0.120 V	10.980 V	65.88 W	0.72 W	98.9%
7.5 A	0.150 V	10.950 V	82.12 W	1.12 W	98.6%
9.0 A	0.180 V	10.920 V	98.28 W	1.62 W	98.4%
10.5 A	0.210 V	10.890 V	114.34 W	2.21 W	98.1%
12.0 A	0.240 V	10.860 V	130.32 W	2.88 W	97.8%
13.5 A	0.270 V	10.830 V	146.21 W	3.65 W	97.6%
15.0 A	0.300 V	10.800 V	162.00 W	4.50 W	97.3%
16.5 A	0.330 V	10.770 V	177.70 W	5.45 W	97.0%
18.0 A	0.360 V	10.740 V	193.32 W	6.48 W	96.8%
19.5 A	0.390 V	10.710 V	208.84 W	7.61 W	96.5%
21.0 A	0.420 V	10.680 V	224.28 W	8.82 W	96.2%
22.5 A	0.450 V	10.650 V	239.62 W	10.12 W	95.9%
24.0 A	0.480 V	10.620 V	254.88 W	11.52 W	95.7%
25.5 A	0.510 V	10.590 V	270.05 W	13.01 W	95.4%
27.0 A	0.540 V	10.560 V	285.12 W	14.58 W	95.1%
28.5 A	0.570 V	10.530 V	300.10 W	16.25 W	94.9%
30.0 A	0.600 V	10.500 V	315.00 W	18.00 W	94.6%

* Efficiency at zero current is shown only as a mathematical reference; no useful output power is delivered at zero load.

5. 30 A Performance Result

Parameter	Calculated result
Nominal input voltage	11.1 V
Load current	30 A
Modeled resistance	0.020 Ω
Voltage drop	0.60 V
Loaded pack voltage	10.50 V

Parameter	Calculated result
Nominal electrical power	333 W
Modeled load power	315 W
Resistive power loss	18 W
Modeled efficiency	94.6%

6. Simulation vs. Hardware Results

Parameter	Result	Result type
Pack configuration	3S6P / 18 cells	Design verified
Nominal voltage	11.1 V	Calculated from cell rating
Maximum voltage	12.6 V	Calculated from cell rating
Pack capacity	12 Ah	Calculated from cell capacity
Nominal energy	133.2 Wh	Calculated
No-load pack voltage	≈12.5 V	Photographic hardware observation
No-load current	0.00 A	Photographic hardware observation
Voltage drop @ 30 A	≈0.60 V	Electrical model
Power loss @ 30 A	≈18 W	Electrical model
Load power @ 30 A	≈315 W	Electrical model
Efficiency @ 30 A	≈94.6%	Electrical model

7. Validation Summary

Validation item	Final status
Cell configuration	Verified as 3S6P design
Pack voltage rating	Validated: 11.1 V nominal / 12.6 V maximum
Pack capacity	Validated by configuration: 12 Ah
Nominal energy	Validated by calculation: 133.2 Wh
BMS integration	Verified visually on assembled hardware
Balance connection	Verified visually on assembled hardware
Main power connection	Verified visually on assembled hardware
No-load voltage indication	Verified visually: ≈12.5 V
No-load current indication	Verified visually: 0.00 A
30 A voltage-drop behavior	Validated by 20 mΩ electrical model: ≈0.60 V
30 A power-loss behavior	Validated by I²R model: ≈18 W
30 A efficiency	Validated by model: ≈94.6%

8. Engineering Assessment

The assembled hardware is consistent with the intended 3S6P architecture and the photographed 12.5 V no-load reading is consistent with a charged 3S lithium-ion pack. The 20 mΩ electrical model predicts a 0.60 V sag and 18 W resistive loss at 30 A, corresponding to approximately 94.6% modeled efficiency. The calculated results provide a coherent electrical baseline for the hardware design and are suitable for documenting the design-validation stage of the project.

9. Safety & Test Controls

- Use only appropriately matched and undamaged lithium-ion cells.
- Maintain insulation between cells, bus connections and the enclosure.
- Use a correctly rated BMS for the selected cell configuration and current requirement.
- Use appropriately rated connectors and wiring for the expected current.
- Avoid uncontrolled short-circuit or high-current testing.
- Monitor pack and BMS temperature during sustained high-current operation.
- Use a charger designed for a 3S lithium-ion pack with a 12.6 V charge limit.

10. Final Result

Final design: 3S6P, 18 × 18650 cells, 11.1 V nominal, 12.6 V maximum, 12 Ah capacity, 133.2 Wh nominal energy, 20 mΩ modeled internal resistance, and 3S 40 A BMS protection. The physical pack was assembled successfully, with approximately 12.5 V and 0.00 A observed on the integrated meter under no-load conditions. The modeled 30 A operating point gives approximately 10.5 V loaded voltage, 315 W load power, 18 W resistive loss, and 94.6% efficiency.